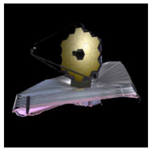

7728 - Investigating a New Evolutionary Channel for Dying Stars

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
pRGB J055102.44-685639.1				
	1	MIRI: J055102.44-685639.1	MIRI Medium Resolution Spectroscopy	(1) J055102.44-685639.1
	6	PRISM: J055102.44-685639.1	NIRSpec Fixed Slit Spectroscopy	(1) J055102.44-685639.1
pRGB star: J051920.18-722522.1				
	2	MIRI: J051920.18-722522.1	MIRI Medium Resolution Spectroscopy	(2) J051920.18-722522.1
	7	PRISM: J051920.18-722522.1	NIRSpec Fixed Slit Spectroscopy	(2) J051920.18-722522.1
pRGB star: J050257.89-665306.3				
	3	MIRI: J050257.89-665306.3	MIRI Medium Resolution Spectroscopy	(3) J050257.89-665306.3
	8	PRISM: J050257.89-665306.3	NIRSpec Fixed Slit Spectroscopy	(3) J050257.89-665306.3
pRGB star: J045755.05-681649.2				

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
	4	MIRI: J045755.05-681649.2	MIRI Medium Resolution Spectroscopy	(4) J045755.05-681649.2
	9	PRISM: J045755.05-681649.2	NIRSpec Fixed Slit Spectroscopy	(4) J045755.05-681649.2
pRGB star: J045555.15-712112.3				
	5	MIRI: J045555.15-712112.3	MIRI Medium Resolution Spectroscopy	(5) J045555.15-712112.3
	10	PRISM: J045555.15-712112.3	NIRSpec Fixed Slit Spectroscopy	(5) J045555.15-712112.3

ABSTRACT

The demise of most stars in the Universe has traditionally been thought to occur as a result of heavy mass-loss (with rates as high as 1e-4 Msun/yr) on the Asymptotic Giant Branch (AGB), when the stars are very luminous (5000-10,000 Lsun) and cool (Teff < 3000 K). However, recently, an exciting new evolutionary channel has been identified from a study of evolved stars in the Magellanic Clouds (MCs). This channel operates when the primary star is much less luminous, i.e., still on the Red Giant Branch (or RGB), likely as a result of a strong binary interaction. The most prominent Galactic representative of this process is the Boomerang Nebula (also the coldest object in the Universe): in it the mass-loss that stripped the primary star of its envelope was driven at an extreme rate (1e-3 Msun/yr) and expansion velocity (165 km/s), much higher than found for any AGB star, and occurred over a relatively short period (3500 yr). The binary companion most likely merged with the core of the primary, producing a jet-driving central dusty disk/torus around the merged object. We propose a JWST study of a select sample of post-RGB objects in the LMC to obtain 5-28 micron spectra that are vital for probing the properties of the mass-ejecta and thus constrain the ejection process . Our results will help constrain existing theoretical models of strong binary interactions for producing post-RGB objects; important diagnostics include the mass of material ejected in the equatorial plane, compared to that in a shell, and the chemical composition of the ejecta.

OBSERVING DESCRIPTION

We propose 19.5 hours of JWST time to investigate the near and mid-infrared dust excesses that have been found around a sample of post-RGB stars in the LMC. The evolution of very luminous (L ~ 6000 Lsun) AGB stars is controlled by heavy mass-loss via dusty winds, thus mid-IR excesses are to be expected. However, a new class of lower luminosity, post-RGB stars (L < ~ 1000 Lsun) that also exhibit dust excesses have recently been identified in the LMC and SMC using optical spectroscopy and Spitzer photometry. For these stars, strong interactions with a compact binary companion during the RGB phase is believed to remove most of the primary's stellar envelope, but with a significant fraction of the ejecta still residing in a disk. Existing theoretical models of strong binary interactions for producing pRGB objects such as CE evolution, or grazing envelope ejection (GEE) are still in their infancy. The focus of this proposal is to investigate this exciting new evolutionary channel for producing proto-planetary nebulae (PPNe) that REQUIRES binary interaction, but which occurs when the primary star is still on the Red Giant Branch (or RGB). Detailed observational studies of this class of objects to elucidate the properties of their mass-ejecta will provide much-needed constraints for theoretical efforts to understand binary stellar interactions.

We propose to determine the physical and chemical properties of the mass ejecta of 5 post-RGB stars in the LMC by using NIRSpec/PRISM and MIRI/MRS to obtain 0.6-28 micron spectra of the sources. The SEDs of our target post-RGBs were modeled using the 1-D radiative transfer code DUSTY, all of which show significant near and mid-IR excess. The modeled SEDs were used as inputs to the JWST ETC to estimate the exposure times and SNR.

Since our objects are all expected to be point sources in the near-IR, we elect to not carry out target acquisition and instead rely on JWST's pointing using the Gaia DR2 coordinates of the guide stars to place the objects within the field. We use NIRSpec in the low resolution mode (PRISM) where we can obtain complete wavelength coverage from 0.6 to 5.3 microns. This region is sensitive to thermal emission from hot dust, scattered light from the central star, and may also show weak molecular bands due to, e.g., H2O and CO, arising in the inner regions of the disk. We adjusted our observations to obtain a SNR ~ 100 in this mode. This resulted in groups/integration ranging from 10 for J055102.44-685639.1 to 35 for J045755.05-681649.2. We use the S200A1 slit and corresponding subarray (SUBS200A1) and NRSRAPID readout pattern.

For MIRI, we will use TA as the sources may be extended. We use all three grating positions for MRS spectroscopy and the FAST readout pattern throughout. The exposure times (number of groups and number of dithers) were decided based on the considerations that we achieve a SNR ~ 20 at 5 microns and a SNR ~ 5.0 or greater at 22 microns. The region of 5-20 microns is especially crucial to understanding the chemistry and spectral characteristics of our sources as it is the region of the 10 and 18 micron silicate emission and also the region of prominent PAH emission. J045755.05-681649.2 is the faintest target in our list; even for this object, we achieve a SNR ~ 20 at 5 microns in a modest amount of exposure time. The SNR is greater at longer wavelengths thanks to silicate emission and circumstellar dust, with the maximum being > 110 at 10 microns. For this source we used 360 groups per integration with a 4-point dither to gain the required amount of exposure time while keeping individual integrations below 1000 s. J050257.89-665306.3 is the next faintest, and we have also chosen a 4-point dither for it with approximately 700 s per integration. For the other three sources we have chosen 2-point dithers with appropriate groups per integration to achieve the required sensitivity; 2-point dithers are acceptable since we are not concerned with PSF oversampling.

Because these are relatively crowded fields, we have elected to do simultaneous imaging at 7.7 microns (F770W) to allow for accurate astrometric reconstruction. 20 groups per integration were selected to allow SNR ~ 10 on 17th magnitude (Vega) WISE W1 sources (of which there are many in the imaging field) but still allow usable information on 12th magnitude WISE W1 sources.

The proposed targets and their fundamental properties as derived from modelig the photometric SED are tabulated below.

Object	Teff	Class	Inner disk					Outer shell					Lum.
			(Td	amin	amax	Y	Mgd)	(Td	amin	amax	Y	Mgd)	
	(K)		(K	micr	micro		Msun)	(K	micr	micr		Msun)	Lsun
J055102	7625	disk	2000	0.005	0.05	7.0	1.99e-8	350	0.005	0.07	3.0	3.05e-3	621
J051920	4500	shell	500	0.3	20.0	2.0	2.59e-5	110	0.005	0.25	20.0	3.44e-2	582
J050257	4586	disk	1200	0.3	5.0	2.0	5.77e-8	250	0.005	1.0	10.0	2.68e-4	303
J045755	4927	disk	1300	0.005	2.0	2.0	9.64e-9	400	0.10	1.0	30.0	5.73e-5	217
J045555	10000	disk	800	0.005	0.25	5.0	2.67e-6	500	0.005	0.25	2.0	8. 73e-5	621

Teff: effective temperature

Td: dust temperature at shell inner radius

amin: min. grain size for model

amax: max. grain size for model

Y: ratio of the dust shell's (disk's) outer to inner radius

Mgd: (inferred) circumstellar (gas+dust) mass; Lum.: (inferred) luminosity

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	J055102.44-685639.1	RA: 05 51 2.4500 (87.7602083d) Dec: -68 56 39.01 (-68.94417d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0	
Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB. Jmag=16.399 *Category=Star* *Description=[Post-asymptotic giant branch]*				
(2)	J051920.18-722522.1	RA: 05 19 20.2600 (79.8344167d) Dec: -72 25 22.26 (-72.42285d) Equinox: J2000	Proper Motion RA: 1.939 mas/yr Proper Motion Dec: 0.964 mas/yr Epoch of Position: 2016.0	
Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB. *Category=Star* *Description=[Post-asymptotic giant branch]*				
(3)	J050257.89-665306.3	RA: 05 02 57.9300 (75.7413750d) Dec: -66 53 6.29 (-66.88508d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0	
Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB. *Category=Star* *Description=[Post-asymptotic giant branch]*				
(4)	J045755.05-681649.2	RA: 04 57 55.1250 (74.4796875d) Dec: -68 16 49.82 (-68.28051d) Equinox: J2000	Proper Motion RA: 1.712 mas/yr Proper Motion Dec: 0.014 mas/yr Epoch of Position: 2016.0	
Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB. *Category=Star* *Description=[Post-asymptotic giant branch]*				
(5)	J045555.15-712112.3	RA: 04 55 55.1670 (73.9798625d) Dec: -71 21 12.17 (-71.35338d) Equinox: J2000	Proper Motion RA: 1.809 mas/yr Proper Motion Dec: -0.731 mas/yr Epoch of Position: 2016.0	
Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB. *Category=Star* *Description=[Post-asymptotic giant branch]*				

Observation

Proposal 7728, Observation 1: MIRI: J055102.44-685639.1

Wed Apr 15 13:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	J055102.44-685639.1	RA: 05 51 2.4500 (87.7602083d)	Proper Motion RA: 0	
		Dec: -68 56 39.01 (-68.94417d)	Proper Motion Dec: 0	
		Equinox: J2000		

Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB. Jmag=16.399
Category=Star
Description=[Post-asymptotic giant branch]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	F560W	FAST	4	1	1	11.1	177419

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	64109
1	SHORT(A)	MRSLONG		FASTR1	70	1	1	Dither 1	4	4	777.011	60851
1	SHORT(A)	MRSSHORT		FASTR1	70	1	1	Dither 1	4	4	777.011	60851
2		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	64109
2	MEDIUM(B)	MRSLONG		FASTR1	70	1	1	Dither 1	4	4	777.011	60851
2	MEDIUM(B)	MRSSHORT		FASTR1	70	1	1	Dither 1	4	4	777.011	60851
3		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	64109
3	LONG(C)	MRSLONG		FASTR1	70	1	1	Dither 1	4	4	777.011	60851
3	LONG(C)	MRSSHORT		FASTR1	70	1	1	Dither 1	4	4	777.011	60851

Observation

Proposal 7728, Observation 6: PRISM: J055102.44-685639.1

Wed Apr 15 13:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	J055102.44-685639.1	RA: 05 51 2.4500 (87.7602083d) Dec: -68 56 39.01 (-68.94417d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0	

Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB. Jmag=16.399
Category=Star
Description=[Post-asymptotic giant branch]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	299074

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	S200A1	NRSRAPID	10	1	1	NONE	3	3	51.475	62977

Observation

Proposal 7728, Observation 2: MIRI: J051920.18-722522.1

Wed Apr 15 13:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	J051920.18-722522.1	RA: 05 19 20.2600 (79.8344167d)	Proper Motion RA: 1.939 mas/yr	
		Dec: -72 25 22.26 (-72.42285d)	Proper Motion Dec: 0.964 mas/yr	
		Equinox: J2000	Epoch of Position: 2016.0	

Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB.
Category=Star
Description=[Post-asymptotic giant branch]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	F560W	FAST	4	1	1	11.1	177436

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	17	1	1	Dither 1	4	4	188.703	64109
1	SHORT(A)	MRSLONG		FASTR1	200	1	1	Dither 1	4	4	2220.032	61926
1	SHORT(A)	MRSSHORT		FASTR1	200	1	1	Dither 1	4	4	2220.032	61926
2		IMAGER	F770W	FASTR1	17	1	1	Dither 1	4	4	188.703	64109
2	MEDIUM(B)	MRSLONG		FASTR1	200	1	1	Dither 1	4	4	2220.032	61926
2	MEDIUM(B)	MRSSHORT		FASTR1	200	1	1	Dither 1	4	4	2220.032	61926
3		IMAGER	F770W	FASTR1	17	1	1	Dither 1	4	4	188.703	64109
3	LONG(C)	MRSLONG		FASTR1	200	1	1	Dither 1	4	4	2220.032	61926
3	LONG(C)	MRSSHORT		FASTR1	200	1	1	Dither 1	4	4	2220.032	61926

Observation

Proposal 7728, Observation 7: PRISM: J051920.18-722522.1

Wed Apr 15 13:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	J051920.18-722522.1	RA: 05 19 20.2600 (79.8344167d)	Proper Motion RA: 1.939 mas/yr	
		Dec: -72 25 22.26 (-72.42285d)	Proper Motion Dec: 0.964 mas/yr	
		Equinox: J2000	Epoch of Position: 2016.0	

Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB.
Category=Star
Description=[Post-asymptotic giant branch]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	299081

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	S200A1	NRSRAPID	15	1	1	NONE	3	3	74.845	62983

Observation

Proposal 7728, Observation 3: MIRI: J050257.89-665306.3

Wed Apr 15 13:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 3:1) Warning (Form): Data Excess over lower threshold

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	J050257.89-665306.3	RA: 05 02 57.9300 (75.7413750d) Dec: -66 53 6.29 (-66.88508d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0	

Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB.
Category=Star
Description=[Post-asymptotic giant branch]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	F560W	FAST	4	1	1	11.1	177444

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	64109
1	SHORT(A)	MRSLONG		FASTR1	250	1	1	Dither 1	4	4	2775.04	61933
1	SHORT(A)	MRSSHORT		FASTR1	250	1	1	Dither 1	4	4	2775.04	61933
2		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	64109
2	MEDIUM(B)	MRSLONG		FASTR1	250	1	1	Dither 1	4	4	2775.04	61933
2	MEDIUM(B)	MRSSHORT		FASTR1	250	1	1	Dither 1	4	4	2775.04	61933
3		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	64109
3	LONG(C)	MRSLONG		FASTR1	250	1	1	Dither 1	4	4	2775.04	61933
3	LONG(C)	MRSSHORT		FASTR1	250	1	1	Dither 1	4	4	2775.04	61933

Observation

Proposal 7728, Observation 8: PRISM: J050257.89-665306.3

Wed Apr 15 13:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	J050257.89-665306.3	RA: 05 02 57.9300 (75.7413750d) Dec: -66 53 6.29 (-66.88508d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0	

Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB.
Category=Star
Description=[Post-asymptotic giant branch]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	299083

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	S200A1	NRSRAPID	20	1	1	NONE	3	3	98.215	62997

Observation

Proposal 7728, Observation 4: MIRI: J045755.05-681649.2

Wed Apr 15 13:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	J045755.05-681649.2	RA: 04 57 55.1250 (74.4796875d)	Proper Motion RA: 1.712 mas/yr	
		Dec: -68 16 49.82 (-68.28051d)	Proper Motion Dec: 0.014 mas/yr	
		Equinox: J2000	Epoch of Position: 2016.0	

Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB.
Category=Star
Description=[Post-asymptotic giant branch]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	F560W	FAST	4	1	1	11.1	177449

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	25	2	1	Dither 1	4	8	566.108	64109
1	SHORT(A)	MRSLONG		SLOWR1	25	2	1	Dither 1	4	8	4873.544	62137
1	SHORT(A)	MRSSHORT		SLOWR1	25	2	1	Dither 1	4	8	4873.544	62137
2		IMAGER	F770W	FASTR1	25	2	1	Dither 1	4	8	566.108	64109
2	MEDIUM(B)	MRSLONG		SLOWR1	25	2	1	Dither 1	4	8	4873.544	62137
2	MEDIUM(B)	MRSSHORT		SLOWR1	25	2	1	Dither 1	4	8	4873.544	62137
3		IMAGER	F770W	FASTR1	25	2	1	Dither 1	4	8	566.108	64109
3	LONG(C)	MRSLONG		SLOWR1	25	2	1	Dither 1	4	8	4873.544	62137
3	LONG(C)	MRSSHORT		SLOWR1	25	2	1	Dither 1	4	8	4873.544	62137

Observation

Proposal 7728, Observation 9: PRISM: J045755.05-681649.2

Wed Apr 15 13:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	J045755.05-681649.2	RA: 04 57 55.1250 (74.4796875d) Dec: -68 16 49.82 (-68.28051d) Equinox: J2000	Proper Motion RA: 1.712 mas/yr Proper Motion Dec: 0.014 mas/yr Epoch of Position: 2016.0	

Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB.
Category=Star
Description=[Post-asymptotic giant branch]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	299084

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	S200A1	NRSRAPID	35	1	1	NONE	3	3	168.325	62998

Observation

Proposal 7728, Observation 5: MIRI: J045555.15-712112.3

Wed Apr 15 13:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	J045555.15-712112.3	RA: 04 55 55.1670 (73.9798625d)	Proper Motion RA: 1.809 mas/yr	
		Dec: -71 21 12.17 (-71.35338d)	Proper Motion Dec: -0.731 mas/yr	
		Equinox: J2000	Epoch of Position: 2016.0	

Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB.
Category=Star
Description=[Post-asymptotic giant branch]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	F560W	FAST	4	1	1	11.1	177458

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	20	2	1	Dither 1	4	8	455.107	64109
1	SHORT(A)	MRSLONG		FASTR1	150	1	1	Dither 1	4	4	1665.024	62259
1	SHORT(A)	MRSSHORT		FASTR1	150	1	1	Dither 1	4	4	1665.024	62259
2		IMAGER	F770W	FASTR1	20	2	1	Dither 1	4	8	455.107	64109
2	MEDIUM(B)	MRSLONG		FASTR1	150	1	1	Dither 1	4	4	1665.024	62259
2	MEDIUM(B)	MRSSHORT		FASTR1	150	1	1	Dither 1	4	4	1665.024	62259
3		IMAGER	F770W	FASTR1	20	2	1	Dither 1	4	8	455.107	64109
3	LONG(C)	MRSLONG		FASTR1	150	1	1	Dither 1	4	4	1665.024	62259
3	LONG(C)	MRSSHORT		FASTR1	150	1	1	Dither 1	4	4	1665.024	62250

Observation

Proposal 7728, Observation 10: PRISM: J045555.15-712112.3

Wed Apr 15 13:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	J045555.15-712112.3	RA: 04 55 55.1670 (73.9798625d) Dec: -71 21 12.17 (-71.35338d) Equinox: J2000	Proper Motion RA: 1.809 mas/yr Proper Motion Dec: -0.731 mas/yr Epoch of Position: 2016.0	

Comments: This is a post-RGB star but since this option is not available in the description selector, we have chosen post-AGB.
Category=Star
Description=[Post-asymptotic giant branch]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	299085

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	S200A1	NRSRAPID	30	1	1	NONE	3	3	144.955	62999